

Subtype readings in Hindi numerical reduplication

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Abstract: Given appropriate context, numerical reduplication in Hindi leads to subtype readings (SRs), where the distribution of the share happens over subtypes of itself. We propose that the use of reduplication in such cases presupposes a partition of the noun set based on a salient property (such as type, or color). This analysis also accounts for non-subtype implicit key cases, such as distribution over implicit individuals or locations. We discuss how SRs reveal the importance of noun set structure (in context) to building distributive meaning, and plural meaning more broadly.

Keywords: *distributivity, dependent indefinites, subtypes*

1 A new key-less distributive meaning

This paper will focus on subtype readings (SRs) arising from numerical reduplication in Hindi, as illustrated by example (1).

- (1) **The Flower Shop Scenario:** A flower shop has many different kinds of flowers: lilies, roses, tulips, chrysanthemums. Ruhi is picking two stems from each kind. The owner says:

Ruhi do-do phool chun rahi hai

Ruhi two-RED flower.PL choose PROG is

‘Ruhi is choosing two flowers of each type’

This usage of numerical reduplication (which is classified as a dependent indefinite (DI), see (Farkas 1997)) leads to distributivity over salient types in the context – in this example, types of flowers.¹ This does *not* have to be strict subtypes, and the distribution can happen over different sizes, colors, and so on. This is illustrated by example (2).

¹Note, however, that while dependent indefinites are often expressed via numerical reduplication cross-linguistically, this does not have to be the case. For example, the Russian dependent indefinite is expressed

- (2) **The Party Store Scenario:** A party store has many different colors of balloons: red, green, yellow, pink, etc. Ruhi is picking two balloons of each color. The owner says:
 Ruhi do-do gubbare chun rahi hai
 Ruhi two-RED balloon.PL choose PROG is
 ‘Ruhi is choosing two balloons of each color’

Generally, the creation of distributive meanings such as the above involves two main parts: the **key** (that which is distributed over), and the **share** (that which is distributed) (Choe 1987, Kuno 1982). Let us take a step back from SRs and consider a more ‘classic’ DI sentence to illustrate the key and share in a concrete example. In (3), the share is ‘children’ and the key is ‘drawings’, with each child having made three drawings. Additionally, DIs are said to induce *covariation* – i.e., DIs require that the values of the key and the share change with one another. In example (3), each child must have made three different drawings (and so ‘drawings’ must take narrow scope with respect to ‘children’).

- (3) BachchoN-ne teen-teen chitra banaaye.
 Children-ERG three-RED picture make.PST
 ‘The children made three drawings each.’

As the reader may have already noted, unlike example (3), the SR cases do not have an explicit key in the sentence – i.e., the key is implicit. The share, flowers/balloons, is distributed over different types/colors of itself (established by context), giving us the subtype reading.

The observation of implicit keys is not new, and is exemplified well in DI languages. Henderson (2014) lays out a typology of such languages based on whether or not they need to be ‘licensed’ by an explicit key (or ‘licensor’) in wide scope. This typology ranges from the requirement of a ‘bona-fide’ quantifier (Russian) or any plural entity (Hungarian, Romanian) in wide scope, to no explicit-key requirement at all (Korean, Telugu, and as we see from examples (1) and (2), Hindi). At the same time, we argue that the SR case described above is different from the implicit key cases that have been noted thus far.

To see why, let us consider the space/time-key readings (terms used by Balusu (2006)) in Korean and Telugu that become salient in the absence of an explicit key. Examples (4a) and (4b) illustrate the distribution of a share (balloons or monkeys respectively) over space and time chunks (Balusu 2006, Choe 1987).

- (4) a. **Korean.** (Choe 1987: 52)
 na-nun phwingsen-hana-ssik-ul sa-ess-ta
 I-TOP balloon-one-DISTR-ACC bought
 ‘I bought one balloon (over time/space)’²
 b. **Telugu.** (Balusu 2006: 3)

via the morpheme *-nibud*, and the Romanian dependent indefinite via the particle *cîte* (Brasoveanu & Farkas 2011, Farkas 1997, Pereltsvaig 2008).

²Choe (1987: 54) did not provide a translation for this example, but mentioned in text that: “ ‘a balloon’ is understood to ‘multiply’ [...] under time, place, etc.”

Ram renDu renDu kootu-lu-ni cuus-ee-Du
 Ram 2 RED monkey-PL-ACC see-PST-3SG
 ‘Ram saw 2 monkeys (in each time interval)’ or ‘Ram saw two monkeys (in each location)’

In both the space/time-key sentences *and* the SR cases, the key is not present in the sentence. However, the SR cases are unique in that the key is not eventive. Further, the key and share are related to one and the same NP, and so the same set of entities in the context.

This first difference is important. Past analyses, based on examples such as (4b), have derived covariation and distributivity via a plurality of events. For example, Balusu (2006) analyzes Telugu dependent indefinites via event partitions. In this analysis, reduplication contributes a D-operator that distributes over an event partition – then, based on whether the subparts are spatial or temporal, we can derive the space/time-key readings as in example (4b). In the presence of a plural or quantified subject, participant-key readings can be derived via a trivial event partition – i.e., an event with no subparts, and a D-operator supplied by the predicate. It is difficult to envision how an event partition analysis could extend to the SR case – an event partition based on subtype does not seem to make sense. In fact, Balusu (2006: 8) himself notes that “a division of the jumping event based on kind or type is not possible”.

Further, a space/time partition *standing* in somehow for subtypes is also not perfect, as subtype structure does not always have to coincide with neat space or time intervals. For example, while we could imagine a situation where Ruhi chose two flowers of each type in neat intervals of time, it is also possible that she walked around the shop, picking one stem from one type before moving onto another bunch and then coming back. Imagine also that the flowers are *not* divided spatially (maybe the owner likes a chaotic design). In this scenario, an event partition based on time or space would not work – yet, the sentence is still true.³

Our second difference points to something important about how subtype/contextual structure can support the creation of distributive and plural meanings. Analyses thus far have focused mainly on the derivation of multiple values for the share variable (i.e., covariation). Based on the language-specific data, this is done in different ways. As we just discussed, Balusu (2006) focused on event partitions to derive space/time-key readings. Henderson (2019), on the other hand, highlights Kaqchikel facts that show that DIs *do not* have to take narrow scope since they can occur with pluractional operators. These operators cannot take scope over simple indefinites (as evidenced by the fact that they do not license covariation of such indefinites), and so the covariation in dependent indefinites cannot be derived as a consequence of narrow scope under a pluractional. This fact drives his dynamic plural logic (DPIL) analysis, which does not rely on the scope of the indefinite. In this account DIs act like normal indefinites but with the addition of a post-supposition that checks for an evaluation plurality in the variable associated with the DI (a plurality arising from different assignments *across* the set of variable assignments, and not a sum in the same assignment). Wide-scope quantifiers thus license DIs via a universal that creates an evaluation plurality of events such that the theta-role dependency between the event parts and the DI variable can hold. Pluractionals do so by creating event and individual subparts (i.e., creating an evaluation

³Thank you to Lydia Newkirk for this line of reasoning!

plurality of events and their participants) and relating these via a theta-role dependency. Note again that this plurality is an evaluation plurality – a plurality across assignments, not a plurality within one assignment.

Crucially for us, it is important that something within the sentence creates a multiplicity of event parts – for example, a wide-scope universal or pluractional. Henderson (2014) has a solution to this for Telugu’s licensor-less space/time-key readings – he proposes that the verb in Telugu may be ambiguous between a single-event and pluractional denotation (being either evaluation singular or plural). If the verb encodes a single-event denotation, we will not get spatial or temporal key readings. If it does, this reading falls out naturally from the semantic composition.

Since this analysis depends on a plurality of events, it cannot extend to our case, where we do not have a licensor that can lead to or contribute such a plurality. It may be the case that the verb in Hindi, too, is ambiguous between an evaluation singular or plural denotation, but as we explained for Balusu (2006)’s partition analysis, the subtype reading is not easily explainable as consisting of a multiple event structure.

Additionally, the fact that the key and share are provided by the *same NP* compels us to look at dependent indefinites through a different lens. It seems that the derivation of multiple values for the indefinite variable is not the only important aspect of DI meaning – context, and the *structure* of the nominal domain (based on subtypes or different colors or sizes) plays an important role in supporting distributive interpretation. This aspect of distributive meaning has not been highlighted by or incorporated into previous analyses. In this paper, we put this property at the forefront via a partition based analysis.

1.1 Partitive

Variation in how the SR cases are expressed supports the analysis that the key and share come from the same contextual NP. Let us consider the same context as we discussed before:

- (5) **The Flower Shop Scenario:** A flower shop has many different kinds of flowers: lilies, roses, tulips, chrysanthemums.
- a. Anu-ne do-do phool chune
Anu-ERG two-RED flower chune.PST
‘Anu ordered two flowers of each type’
 - b. Anu-ne [sare phoolon me-se]_{PP} do-do chune
Anu-ERG [all flowers from] two-RED choose.PST
‘Anu ordered two flowers of each type (from all flowers)’
 - c. Anu-ne [un phoolon me-se]_{PP} do-do chune
Anu-ERG [those flowers from] two-RED choose.PST
‘Anu ordered two flowers of each type (from those flowers)’

The internal partitive⁴ in these examples makes no difference to the meaning. It does, however, serve to make the key (the set of flowers of different types) more explicit. Support for this

⁴Another variation in this construction involves repeating the noun after the numeral: Anu-ERG (those flowers from) two-RED flower choose.PST

comes from the fact that if the specific set or subset is made more explicit in context, the partitive seems redundant:

- (6) a. **Context:** Ruhi has made different kinds of fritters (with different veggies). She sees her friend try one of them and asks her if she tried all the different kinds. The friend replies:
 Haan! Balki, mai-ne do-do kha liye.
 Yes! In.fact, I-ERG two-RED eat do.PST.PL
 ‘Yes, in fact, I had two of each type.’
- b. **Context:** Ruhi has made different kinds of fritters (with different veggies *and* meats). She sees her vegetarian friend try one of them and asks her if she tried all the different veggie ones. The friend can again reply with (6a).

In context (6b), if Ruhi did not know her friend was vegetarian, and asked the question in (6a) instead, her friend could instead reply:

- (7) Haan, mai-ne veggie-waloN se do-do kha liye.
 Yes, I-ERG veggie-ones from two-RED eat do.PST.PL
 ‘Yes, I ate two each of the veggie ones.’

As we will discuss in §4.1, the status of the partitive — whether it is always there for numerical reduplication (either in the subtype case or otherwise) — is an open question.

2 An analysis of the subtype reading: Partitions in context

It is clear that we need a way to involve *context* in our account of SRs. In order to talk about subtypes, there must be a certain structure in the context that *allows* for subtypes in the first place (like the flower or balloon shop scenarios). This structure in context, and the access to it, will be central to our analysis.

2.1 Plurality & dependence on subtypes

We first note that the role of subtypes in the interpretation of plural expressions has recently been observed in the literature. Dayal, Sánchez & Vengoa (in prep) offer a presuppositional analysis of *kuna*, a plural marker in Cuzco Quechua, which, similarly to Hindi, has the reliance on subtypes as one of its properties. For example, in (8a), we see that *kuna* creates reference to subtypes of guinea pigs. Without *kuna* in (8a), there would be no reference to subtypes. Dayal, Sánchez & Vengoa (in prep)’s “sorting mechanism hypothesis” argues that *kuna* presupposes some way of partitioning the set of individuals in the domain that are associated with the *kuna* marked noun. This is also supported by the fact that *kuna* can be used with a noun like *dinosaur*, which has clear subtypes, but not *dodo* (see also, Dayal to appear).

- (8) a. **Cuzco Quechua.**
 Quwi-kuna-n chiri-pacha-pi wañu-pu-chka-rqa-nku
 guinea-pig-PL-EVID/FOC cold-time-LOC die-reg-PROG-PST-3.PL
 ‘Guinea pigs (of different kinds) kept dying all winter.’

$$b. \llbracket kuna \rrbracket = \lambda P : \exists P \prod_{color/size/shape/type, etc.} (P)(P).P$$

Dayal, Sánchez & Vengoa (in prep) thus make use of a partition along size/color/type for *kuna*'s presupposition — this ensures that *kuna*, when used in a sentence, already presupposes a plural entity that is structured (or is able to be structured) along some salient dimension (acting as a “sorting mechanism”). This analysis points us in a very helpful direction — it seems, like in Quechua, that we need a way of structuring the domain such that the use of numerical reduplication (to give us sub-type readings) is felicitous. Of course, since we are dealing here with numerals, we have to then build in a way to get to the atomic individuals of the correct number too, while that is not needed in the analysis of *kuna*. And, while *kuna* says nothing about distributivity, that is the main import of numerical reduplication. Thus, our denotation should be able to derive that crucial part of the meaning.

2.2 The two parts of meaning for Hindi numerical reduplication

Like Cuzco Quechua's *kuna*, the use of numerical reduplication with a subtype key requires some way to restrict context such that the domain of individuals can be categorized based on a property like size or color. In fact, we can use the very same presupposition as in (8a) in order to achieve this:

$$(9) \quad \exists P \prod_{color/size/shape/type...} (P_i)(P)$$

Like for *-kuna*, this presupposition ensures that there is some contextually-defined way of dividing the set of individuals, P , modified by the numerical reduplication. As we mentioned before, the at-issue content for reduplication then has the job to distribute over the cells of the partition and ensure that a *num* number of items out of each cell are arguments for the main predicate. This can be achieved via the following at-issue content, which uses the contextually salient partition indicated by i :

$$(10) \quad \begin{aligned} a. & \lambda num \lambda P \lambda Q \forall Z [[Z \in \prod (P_i)(P)] \rightarrow \exists y [| \lambda y'.y' \leq y \wedge AT(y')| = num \wedge y \leq \oplus Z \wedge Q(y)]] \\ b. & \forall Z [[Z \in \prod (P_i)(\llbracket FLOWERS \rrbracket)] \rightarrow \exists y [| \lambda y'.y' \leq y \wedge AT(y')| = 2 \wedge y \leq \oplus Z \wedge choose(y)(Ruhi)]] \end{aligned}$$

For the sentence in (1), this would ensure we get the desired meaning outcome: That for all flower cells in the partition ($Z \in \prod (P_i)(\llbracket FLOWERS \rrbracket)$), there exists a sum of 2ys that was chosen by Ruhi!

We have now laid out and briefly demonstrated the two main parts of our account: the presupposition associated with, and the truth conditions of, numerical reduplication. However, the contexts in which this construction can be used are much more varied than just the one example in (1) or (2), including cases of implicit keys that are not subtype-based. It is important that our account works for these contexts as well. We start by exploring our base context further — and then move onto more complicated scenarios.

2.3 Test contexts

2.3.1 The flower shop

Let us set up our world of evaluation: The set of flowers, $P = \{\text{🌻}, \text{🌻}, \text{🌻}, \text{🌸}, \text{🌸}, \text{🌺}, \text{🌺}, \text{🌸}, \text{🌻}, \text{🌺}, \text{🌸}\}$

A possible partition may then be:

$$(11) \quad \Pi_{type}(\mathbb{P})(P) = \{ \{ \text{🌻}, \text{🌻}, \text{🌻}, \text{🌻} \}, \{ \text{🌷}, \text{🌷}, \text{🌷}, \text{🌷} \}, \{ \text{🌹}, \text{🌹}, \text{🌹}, \text{🌹} \} \}$$

Our presupposition – that there exists a way of partitioning according to type (for this example) – passes. Further, our truth denotation gives us the desired result by picking out two flowers from each subtype (each cell of the cover). We can show the final result via the illustration in (12).

(12) Anu ordered 2-2 flowers⁵ $\Rightarrow$ + + + + +

2.3.2 Judging a competition

- (13) **Competition scenario:** A competition where participants had to grow 3 flowers, of any kind. The judge chooses 2 from each participant to judge.

While this context may seem like it does not fit into the paradigm we have been building, we see that it is not difficult to mold the presupposition to our purposes – if we consider the appropriate context. Our partition, defined based on a contextual property (belonging to a participant), would look like (14).

$$(14) \quad \prod_{participant}(\mathbb{P})(P) = \{ \{ \text{🌻}, \text{🌻}, \text{🌹} \}, \{ \text{🌷}, \text{🌷}, \text{🌷} \}, \{ \text{🌻}, \text{🌷}, \text{🌹} \} \}$$

Once again, our presupposition – although here based on a more ad-hoc property – is satisfied. The denotation, which then allows quantification over the cells of the partition in (14) will give us a result like the one in (15):

(15) The judge chose 2-2 flowers $\Rightarrow$  ,  +  ,  +  , 

This example shows us that the property over which the partition is defined can be pretty broad — it is not limited to subtype, color, or size. This highlights the benefit of using a partition to achieve our result over something like taxonomic hierarchies, which would be more limited. Conversely, the fact that numerical reduplication can be used in contexts like this points to something like partitions being in play as opposed to kinds or taxonomies only.

Finally, this Hindi example is reminiscent of the German ‘je’ example in (16) also previously discussed as evidencing implicit keys. This example refers to three rotten apples (the share) coming from each basket in the context (the key).

- (16) German. Link (1987: 175) & Choe (1987: 63)
 Je drei Äpfel waren faul
 DISTR three apples were rotten
 'Three apples each were rotten'

⁵We use the English translation sentence for ease of illustration.

This parallel shows that our analysis can extend to cases discussed previously – and to cases where the key is implicit *and* external to the modified NP (i.e., not coming from the same set as the share). Note that these cases *may* coincide with ‘spatial’ or ‘temporal’ keys. We discussed that SRs do not have to overlap perfectly with spatial/temporal intervals, but this does not mean that they cannot. For example, the participants in example (15) could be in different locations – giving us an overlap with a space-based partition.

2.3.3 Single type in context

So far, we have seen three contexts (and partitions) that our analysis works for. Let us now consider a context that we would *not* want our presupposition to allow for, the partition in (17).

$$(17) \quad \prod_? P(P) = \{ \{ \text{🌻}, \text{🌻}, \text{🌻}, \text{🌻}, \text{🌻} \} \}$$

This context has only a single type of flower, and thus the use of reduplication in this situation should be infelicitous. This situation is appropriately ruled out by our partition presupposition – there is no way to divide our noun set into parts, i.e., the partition does not exist.

Additionally, this situation is also ruled out by economy considerations. Once we take a look at the result of our truth conditions, this gets a bit clearer.

$$(18) \quad \text{Anu ordered two-two flowers} \longrightarrow \text{🌻} + \text{🌻}$$

This is a result that is equivalent to the sentence in (19), with an unreduplicated numeral.

$$(19) \quad \text{Anu ordered two flowers} \longrightarrow \text{🌻} + \text{🌻}$$

When a speaker chooses to use a reduplicated numeral, they are choosing a more morphologically complex construction. This morphological complexity must be justified, and the simple construction would be preferred due to economy constraints. Thus, the moment a speaker chooses to use reduplication, this choice *already* signals proper variation with *multiple* types. In all other cases *except* the single type case, we in fact, do get different results via the simple numeral vs. reduplication – meaning that this consideration will apply only to the single type case we want to rule out.

We can also extend this explanation to cases where each cell of the partition includes an amount of entities equivalent to the number being reduplicated – such as 2 flowers of each type when using 2-2. Since the reduplication here would amount to covering *everything* in the partition, the speaker could simply use the word for ‘all’ instead. This is appropriate since such a context would not fully license the use of 2-2.

2.4 Including the partitive

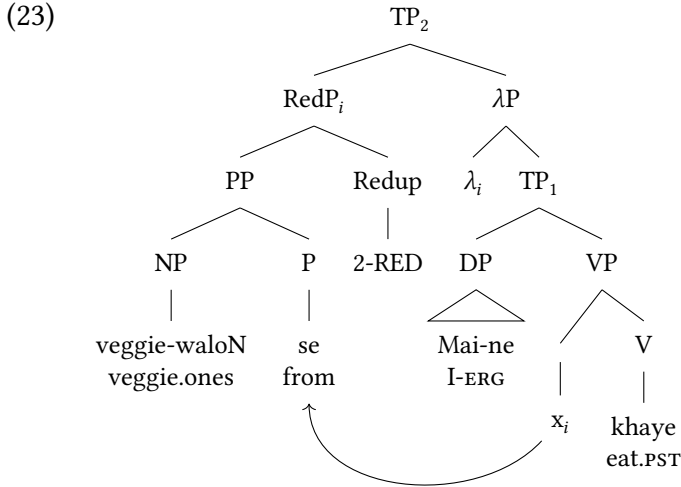
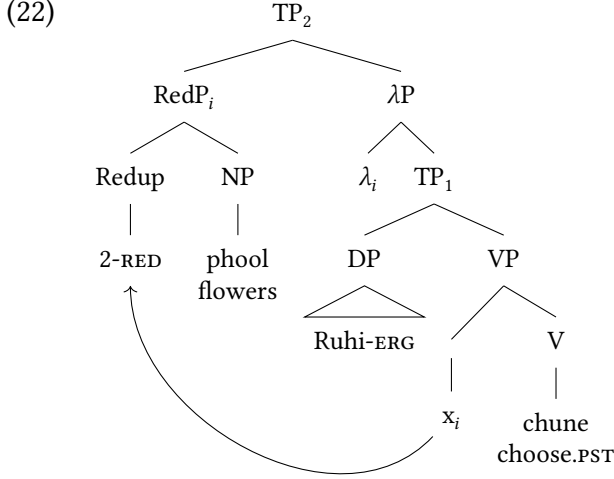
The partitive can be included easily into our analysis. Instead of directly modifying the noun, the reduplication-number complex combines with the inner partitive, giving us the following:

$$(20) \quad \forall Z \left[[Z \in \prod(P_i)(\lambda x.x \leq \llbracket \text{VEGGIE.ONES} \rrbracket)] \rightarrow \exists y \left[[\lambda y'.y' \leq y \wedge AT(y')] = 2 \wedge y \leq \oplus Z \wedge \text{ate}(y)(I)] \right] \right]$$

This allows us to, for example, exclude a type in context if we want to (say Anu is allergic to sunflowers – the partitive inside the partition would allow us to exclude this type, i.e., x would no longer include sunflowers). We assume that the partitive has the following denotation, in line with previous literature:

$$(21) \quad \lambda y. \lambda x. x \leq y$$

Representative trees for both versions are as follows:



3 Cross-linguistic examples: the prevalence of subtype keys

In addition to Hindi, we have confirmed SRs in languages previously discussed in the DI literature. This includes Telugu and Hungarian, along with Yoruba. The prevalence of SRs again indicates that context and nominal domain structure forms an important basis for the creation of distributive meanings, and points to something fundamental about distributivity/plurality phenomena.

- (24) a. Sreekar Raghotham (p.c.)
Telugu. Newspaper context: Seema gets four newspapers delivered to her house every day. Two English, and two Telugu. She says:
 memu rendu-rendu newspaper-lu teppincukuntaamu.
 we two-RED newspaper-PL get.delivered
 ‘We get two papers delivered of each language.’
- b. Gábor Alberti (p.c, based on Dékány & Csirmaz (2017), pp. 1082–1083)
Hungarian. Flower Shop Context (same as Hindi examples):
 Robi két-két virágot vett.
 Robi two-RED flower.PL buy.PST
 ‘Robi bought two flowers of each type.’
- c. Olabode Adedeji (p.c.)
Yoruba. Flower Shop Context (same as Hindi examples):
 Ruhi ra òdòdó méjì-méjì
 Ruhi buy flower two-RED
 ‘Ruhi bought two flowers of each type’

Outside of DIs, SRs have been noted for ‘X-woo-X’ constructions in Mandinka (Cisse & Coppock 2024). Here too, we see distributivity over different subtypes in context. For example, a sentence like (25) is generally not felicitous if everything is carried at once (same time and same space). If, however, the eggs are of different types, this sentence is okay. Thus, in Mandinka too, we see spatial/temporal *and* subtype keys.

- (25) Mandinka (Cisse & Coppock 2024: 80)
 Fode ye siise-e kili-woo-kili samba le
 Fode PRED chicken egg-DISTR-egg carry PERF
 ‘Fode carried each chicken egg’

Finally, we can mention the well-known SRs arising from the pluralization of mass nouns.

- (26) Mass Nouns.
 That bar carries her favorite beers.

4 Discussion

We have now introduced SRs in Hindi (and a few other languages), and described how they add to the distributivity literature. First, SRs add to the typology of possible keys. Second, we have argued that SRs point to the importance of contextual structure, and that this structure may even subsume some aspect of eventive keys, such as space-based distributivity in the German example in (16). However, we have thus far avoided the more ‘classic’ cases in the distributivity literature, where the key is explicit in the sentence. What the reader might be wondering is how this analysis extends to these classic cases. A number of analyses have already been proposed to account for them. We discuss two examples (with a simple plural, and a universal quantifier) to show how ours can too. Please note that, for all cases discussed

below, we follow and rely heavily on Balusu (2006)'s analysis of these cases in the context of his event-partition analysis. We end with a discussion of an 'each and every' interpretation associated with a focused use of reduplicated 'one'.

4.1 The elephant in the room

Note that bringing in 'the classic case' will bring in the possibility of 'double distributivity' (also discussed by Balusu (2006)) – i.e., distribution over both the explicit key, *and* an implicit one (like subtypes). We will discuss these cases below, along with the usual single-level distribution ones.

4.1.1 Plural

Example:

- (27) BachchoN-ne do-do filmeN dekhiN
 Children two-RED film.PL see.PST
 'The children watched two films each'

1. Classic case

For this example, the classic case would involve the children watching two films each. This reading can be accounted for following a strategy similar to Balusu (2006). Balusu (2006) proposes that a D-operator associated with the predicate distributes over the plural entity, and the D-operator associated with the reduplication combines with a *trivial* event partition (for us, a trivial partition over individuals).

However, perhaps the reduplication contributes, optionally, a D-operator itself. This would explain why in sentences like (28), reduplication *needs* some type of key, be it individuals or subtypes. Without either, the sentence is not felicitous. If the D-operator was simply associated with the predicate, then no D-operator plus a trivial partition could allow **no distributivity**. Balusu explains this via his plurality requirement – which states that in a sentence like (28), the number of films watched must be greater than 2. This ensures 'covariation,' the requirement that the share must be different across the different key participants. However, this plurality requirement does not explain the *enforcement* of a D-operator associated with the predicate, i.e., how the reduplication places a restriction on the denotation of the predicate.

2. Double distributivity

For double distributive readings, consider a case where there is a collection of films divided by genre. A 'double distributive' reading would be one where each child watched two films of each genre. We can account for this case via the optional D-operator *and* a non-trivial partition (as in Balusu (2006)).

3. Collective

Finally, a collective reading, where the kids together watched two films of each type, would be related to the absence of the optional D-operator in combination with a partition based on genres, giving us distribution over types but not participants.

4.1.2 Quantifier

Example:

- (28) Har bachche-ne do-do filmeN dekhiN
 Every child-ERG two-RED film.PL see.PST
 'Every child watched two films'

1. Classic case

The fact that the quantifier can co-occur with the reduplication presents a redundancy (as noted also by Balusu (2006)). The quantifier contributes its own universal, which can distribute over the kids, and the D-operator from the reduplication is not needed. Like for the plural, a trivial partition in combination with this D-operator could give us the 'classic,' non-subtype, reading.

2. Double distributivity

Again, we face the same redundancy. Double distributivity may arise the same way as for the plural, except that the D-operator contributed by the reduplication is not needed (or is redundant). We already have the universal from the quantifier, and all we need is a non-trivial partition, like for the plural double distributive case.

3. Collective

There is no collective reading for this sentence!

There are two issues that stand, however, in the way of a unified analysis for the classic case and the subtype readings. 1) What is the status of the inner partitive?, and 2) What is the status of the presupposition? For 1), the subtype readings could be a subcase of the overt partitive – i.e., the partitive could be present in all cases where a subtype/implicit key reading arises. For our judge example, we could imagine a sentence like (29), with the inner partitive phrase being covertly or overtly present.

- (29) Judge-ne (logon-se) do-do phool chune
 Judge-ERG people-from two-RED flowers choose.PST
 'The judge chose two flowers from each participant'

However, the partitive does not, at first glance, make sense for 'the classic case.' The same situation arises for the latter issue. We need the presupposition for SR cases, but there is no infelicity when sub-types readings are not present – which is unexpected given the predicted presupposition failure. The question of the unification of these analyses thus remains an open question.

4.2 Focused readings

Recall that we previously ruled out the use of reduplication with a singular subject and single type in context. However, there is actually one case in which this is possible – when a reduplicated 'one' (*ek*) numeral is stressed or focused. For example, the sentence in (30)

— where the bold *ek-ek* points to focus — is perfectly acceptable with just one type in context. This focused use, which leads to an ‘each and every’ interpretation, is limited to the reduplication of *one*.

- (30) Ruhi-ne **ek-ek** phool khareed liya.
 Ruhi-ERG one-RED.FOC flower buy do.SG.PST
 ‘Ruhi bought each and every flower’

It was suggested to me by Tom McCoy (p.c.) that perhaps the use of *ek-ek* here coerces the partition into atomic cells like so:

- (31) {{}, {}, {}, {}, {}}

However, the coercion explanation does not make sense in light of the fact that the focused reading is available even when there is a type structure in the context (such as the presence of multiple roses, tulips, sunflowers, etc.). One possibility, suggested to me by Veneeta Dayal (p.c.) is that the focus shifts the attention from the class that the individuals are members of, to the individuals themselves. This then yields a partition with atomic individuals. Questions remain, of course, about how exactly this might be achieved, and the relationship with focus.

5 Conclusion

In this paper, we provided an analysis for subtype readings associated with numerical reduplication in Hindi. We discussed how these readings were unique in that the key and share are related to the same noun set (with the share being distributed over subtypes of itself). Our partition-based analysis accounts for these readings, along with other implicit-key cases (for example, when the distribution is over locations), via a sortability presupposition. Subtype readings point to the importance of context, and the structure of the nominal domain in context, to the building of distributive meaning. In fact, the prevalence of subtype-based interpretations in a variety of plural meaning phenomena (such as X-woo-X constructions in Mandinka, the plural marker in Cuzco Quechua, and mass nouns) points to something broader about the contribution of subtype structure to plural meaning.

Abbreviations

3 = third person, ACC = accusative, DISTR = distributive, ERG = ergative, FOC = focus, LOC = locative, PERF = perfective, PL = plural, PRED = predicative, PROG = progressive, PST = past, RED = reduplicant, SG = singular, TOP = topic.

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